HEALTHCARE ANALYTICS KPI VALIDATION EVIDENCE

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ANALYSTLAB AFRICA

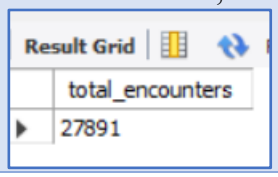
KPI VALIDATION IN MYSQL

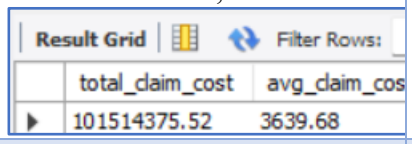
Purpose of Validation

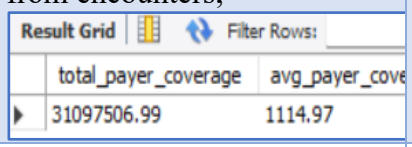
To ensure the accuracy of dashboard calculations, key metrics displayed in Power BI were independently validated using SQL queries executed against the MySQL database. This process ensured consistency between the database and the report.

Metrics Validated

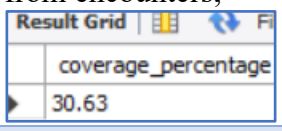
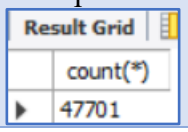
The following metrics were validated using MySQL:

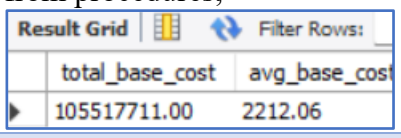
Metrics	Power BI Calculation Results	MySQL Validation Results	Match ? Yes/No
Total Encounters	`Total Encounters = DISTINCTCOUNT('hospital_analytics encounters'[encounter_id])` **Result = 27,891**	select count(distinct encounter_id) as total_encounters from encounters; 	Yes

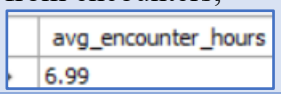
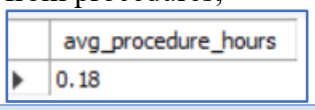
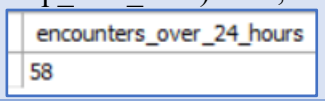
Total and Average Claim Cost	`Total Claim Cost = SUM('hospital_analytics encounters'[total_claim_cost])`
	Result = $101.51M
	`Average Claim Cost = AVERAGE('hospital_analytics encounters'[total_claim_cost])`
	Result = $3.64K
		select sum(total_claim_cost) as total_claim_cost, round(avg(total_claim_cost), 2) as avg_claim_cost from encounters;
		 
			Yes

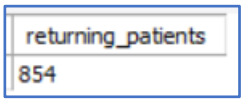
Total and Average Payer Coverage	`Total Payer Coverage = SUM('hospital_analytics encounters'[payer_coverage])`
	Result = $31.10M
	`Average Payer Coverage = AVERAGE('hospital_analytics encounters'[payer_coverage])`
	Result = $1.11K
		select sum(payer_coverage) as total_payer_coverage, round(avg(payer_coverage),2) as avg_payer_coverage from encounters;
		 
			Yes

Coverage Percentage	`Coverage Percentage = DIVIDE([Total Payer Coverage],[Total Claim Cost])` Result = 30.63%	select round(sum(payer_coverage) / sum(total_claim_cost) *	Yes

		100,2) as coverage_percentage from encounters; 	
Total Procedures	`Total Procedures = COUNT('hospital_analytics procedures'[patient_id])` Result = 47,701	select count(*) from procedures; 	Yes

Total and Average Procedure Base Cost	`Total Procedure Base Cost = SUM('hospital_analytics procedures'[base_cost])`
	Result = $105.52M
	`Average Procedure Base Cost = AVERAGE('hospital_analytics procedures'[base_cost])`
	Result = $2.21K
		select sum(base_cost) as total_base_cost, round(avg(base_cost),2) as avg_base_cost from procedures;
		 
			Yes

Average Encounter Duration in Hours	`Average Encounter Hours = AVERAGE('hospital_analytics encounters'[duration_hours])` Result = 7.27	select round(avg(timestampdiff(hour, start_date_time, stop_date_time)),2) as avg_encounter_hours from encounters; 	No
Average Procedure Duration in Hours	`Average Procedure Hours = AVERAGE('hospital_analytics procedures'[duration_hours])` Result = 0.16	select round(avg(timestampdiff(hour, start_date_time, stop_date_time)),2) as avg_procedure_hours from procedures; 	No
Encounters longer than 24 hours	`Encounters Longer than 24 Hours = CALCULATE( COUNTROWS('hospital_analytics encounters'), 'hospital_analytics encounters'[duration_hours] > 24 )` Result = 75	select count(*) as encounters_over_24_hours from encounters where timestampdiff(hour, start_date_time, stop_date_time) > 24; 	No

Returning patients	Returning Patients =
	COUNTROWS(
	FILTER(
	VALUES('hospital_analytics encounters'[patient_id]),
	CALCULATE(COUNTROWS('hospital_analytics encounters')) > 1
	)
	)
	Result = 854
		select patient_id,
		count(encounter_id)
		from encounters
		group by patient_id
		having count(patient_id) > 1;
		
		-- count returning patients
		select count(*)
		from (select patient_id
		from encounters
		group by patient_id
		having
		count(patient_id) > 1) t;
		
		 
		
			Yes

Patient Return Rate	Patient Return Rate = DIVIDE([Returning Patients],[Total Patients]) Result = 87.68%	select round((select COUNT(*) from (select patient_id from encounters group by patient_id having COUNT(encounter_id) > 1) t) / COUNT(distinct patient_id) * 100, 2) as return_rate from encounters;  	Yes

Validation Results

Most of the Power BI measures produced results that matched SQL calculations. The consistency between platforms confirms the reliability of the data model, DAX calculations, and dashboard outputs.

Duration mismatch

Power BI’s calculations were done using fractional hours while SQL’s calculations used the whole timestamp to make its calculations. This may have resulted in the differences in answers.

Further validation is required to match the numbers.